COLLEGE 117

MARCH 30, 2026

PROF. DAPHNE MARTSCHENKO

PROF. NICOLE MARTINEZ



Welcome to COLLEGE 117

Just Biotechnologies: Who Benefits and Who Is Left Behind

About us

Prof. Nicole Martinez-Martin
nicolemz@stanford.edu



- Stanford Center for Biomedical Ethics
- Background: Law & Anthropology
- Research: Ethics of Medical AI, Neuroethics

About us

Prof. Daphne Martschenko
daphnem@stanford.edu



- Stanford Center for Biomedical Ethics
- Background: Education & Medical Anthropology
- Research: Ethics of Genetics

About us

Dr. Meghan Warner



- COLLEGE
  (Civic, Liberal, and Global Education)
- Background: Sociology
- Research: Birth; Sexual Violence

mowarner@stanford.edu

About us

Dr. Richard McGrail



- PhD, Anthropology, Stanford Univ., 2024
- Research: Foster Care | Culture & Adolescent Mental Health
- Hobbies: Photography | Woodworking | Outdoors | Fun
- Featured in Image: Roscoe 🐶
- Email: mcgrail@stanford.edu

Laptop Policy

Laptops, tablets and phones (as well as headphones/earbuds) should be put away and not used during lectures unless you have an accommodation requiring one

Course “housekeeping”

Read the syllabus

- Communication is important – be in contact teaching fellows about course issues/questions
- Although the answer to your question might be in the syllabus, so start by reading it

OAE Letters

- Send OAE letters to your section leader
  - **mowarner@stanford.edu – Dr. Meghan Warner**
  - mcgrail@stanford.edu – Dri. Richard McGrail

Student Athletes

- Be sure to be in contact (beforehand) with your section leader about each absence from lecture/section

Course “housekeeping”

Student Hours

- Valuable for building relationships with folks who you may one day want to write you a reference!

Course “housekeeping”

Absences

- Excused absences: unavoidable, such as illness, athletic event out-of-town, representing university
  - Limits on # of excused absences from section and lecture
  - You can make up the credit for an excused absence
  - Need to email your teaching fellow ahead of time for planned absences and within reasonable time for unplanned

Lecture attendance

- Pollev attendance
  - Students are expected to be on-time, prepared for full lecture and section times

Overview of Course

Biotechnologies

Genomics

Biotechnologies

Genomics

AI

Biotechnologies

Genomics

AI

Neurotechnology

Exploring: What is justice?

Who benefits from advances in science and technology, and who is excluded?

Is research and development in genomics, AI, and neurotechnology proceeding in ways that are inclusive of a wide set of perspectives and populations?

Exploring: What is justice?

"Who ought to receive the benefits of research and bear its burdens?"
(Belmont Report)

Distribution of burdens and benefits:

- to each person an equal share
- to each person according to individual need
- to each person according to individual effort
- to each person according to societal contribution
- to each person according to merit

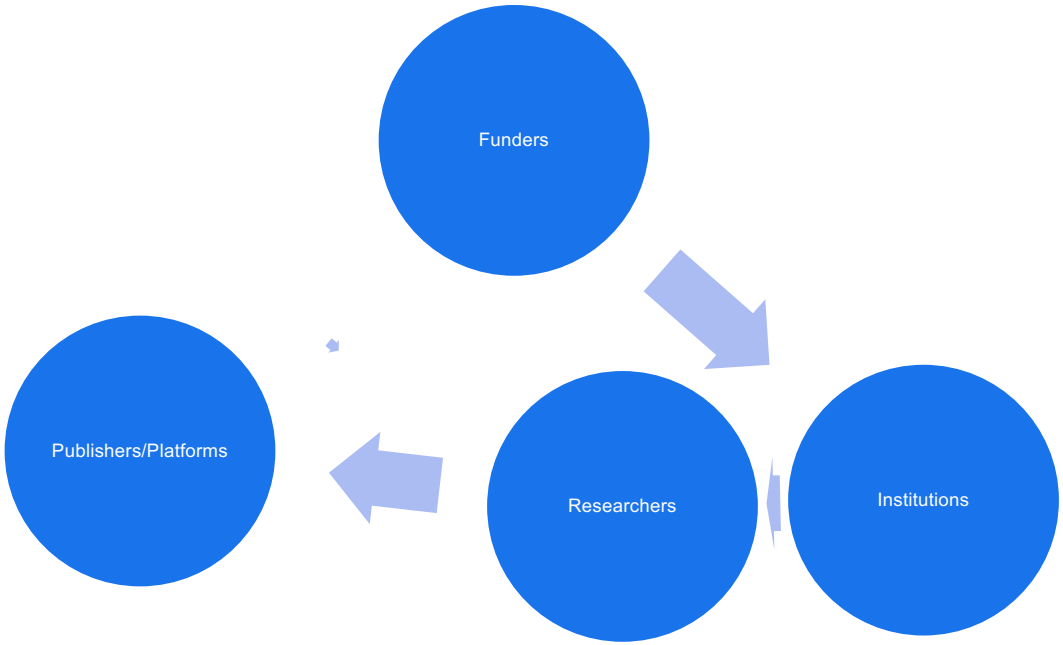
Broadly: Giving each person their due

- **What social structures support justice in society?**
  What processes?
  What liberties/rights are needed?

Course Learning Goals

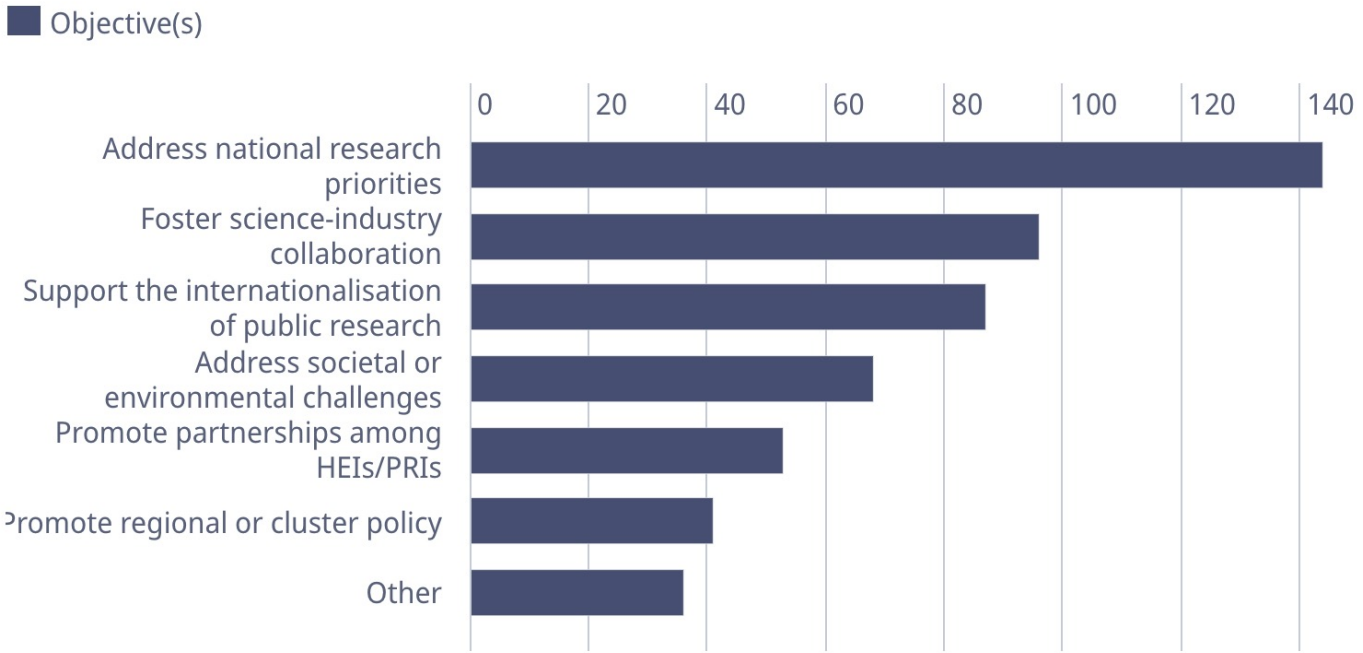
1. Evaluate how existing social arrangements create and maintain justice issues, in different global contexts, in relation to genomics, medical AI, and neurotechnologies.

Scientific Research Infrastructure

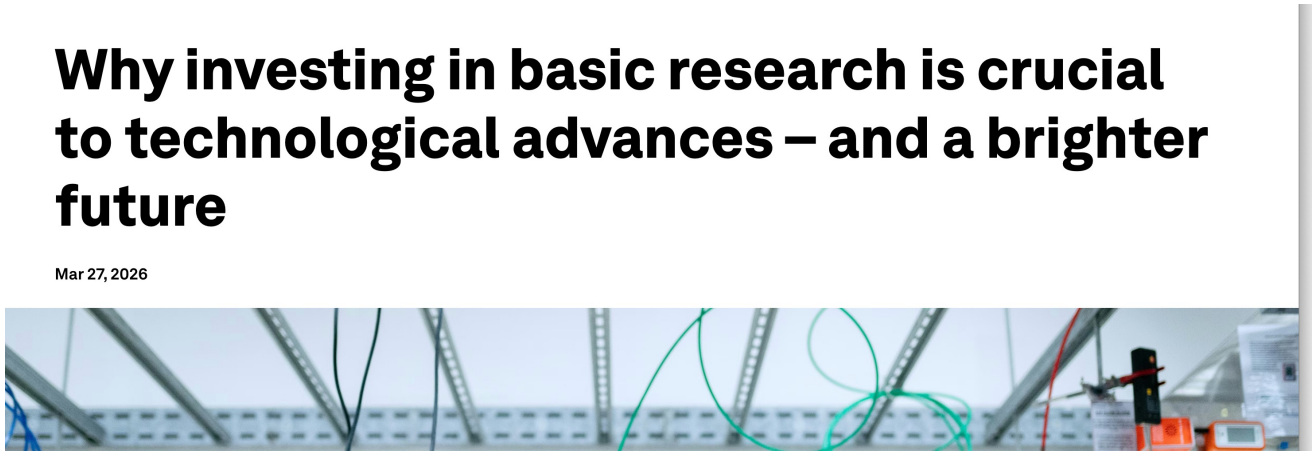


Objective(s) of the policies for research and technical infrastructures

Number of objectives



Source: EC-OECD (2024), STIP Compass: International Database on Science, Technology and Innovation Policy (STIP), 6 February 2024 edition, https://stip.oecd.org



Example: NIH Funding

- **Trains and funds scientists**
- **Basic science vs. specific applications**
- **99.4% of FDA drugs were funded by NIH grants**
- **Develops technologies: pharmaceuticals, AI, medical devices etc.**

Course Learning Goals

1. Evaluate how existing social arrangements create and maintain justice issues, in different global contexts, in relation to genomics, medical AI, and neurotechnologies.

2. Discuss biomedical research and technologies as part of knowledge systems that reflect and can contribute to societal injustices.

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3. Evaluate and apply different ethical frameworks for assessing justice issues in global health technologies.

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1. Evaluate how existing social arrangements create and maintain justice issues, in different global contexts, in relation to genomics, medical AI, and neurotechnologies.

2. Discuss biomedical research and technologies as part of knowledge systems that reflect and can contribute to societal injustices.

3. Evaluate and apply different ethical frameworks for assessing justice issues in global health technologies.

4. Compare different countries' approaches to addressing ethical development of and justice in genomics, medical AI, and neurotechnologies.

What infrastructure & resources ethical development and deployment of research and health technologies in countries?

- Who decides what is ethical research?
  - E.g., Institutional Review Boards
- Who decides what is a “harm”?
  - Individual vs. community or societal harms

Assignments

Assignment 1: 10% - **Small Group Activity & reflection**

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Assignment 2: 20% - **Application of Justice analysis to media piece**

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Assignment 2: 20% - Application of Justice analysis to media piece

Assignment 3: 25% - Unessay – engage with justice issues for a selected technology and global region

Week 1 - Introduction to Ethical Frameworks



Veil of Ignorance